



ON THE FACTORY FLOOR

At first, airframes were mostly made using hand tools, often in workshops employing no more than a few dozen staff, and the production process was slow and laborious. More mechanized, larger-scale production only developed after 1911.

THE FLYING DUCK

On March 28, 1910, 28-year-old French engineer Henri Fabre made the first takeoff from water in his canard (tail-first) seaplane, the Hydravion, at Lake Berre near Martigues on the Mediterranean, despite never having flown before. The Hydravion had three floats and was powered by a 50hp Gnome engine.

From 1909 onward, there was a swift expansion in the range of aircraft designs. While great strides were being made in the new science of aerodynamics, it was not until World War I that such insights would seriously affect practical aircraft design. Successful airplanes evolved instead through the accumulation of a practical body of knowledge based on the experience of flying and of building flying machines. Small-scale manufacturing companies, often set up by experienced pilots, employed engineers and artisans who might

have previously worked on anything from ship-building to furniture making. The production process was slow and laborious. More mechanized, larger-scale production only began to develop after 1911, when the first military contracts arrived.

Production line

The French led the way in aircraft production – France's lead in the conquest of the air became an important focus of national pride. By 1910, companies run by Farman, the Voisin brothers, Levavasseur, and Blériot had staked out the

territory for which others had to compete. They were joined in 1910–11 by several other notable names. The Caudron brothers, Gaston and René, based in Picardie, became famous for their distinctive tractor biplanes. Louis Bréguet, descended from a family of wealthy Parisian clockmakers, opened a factory in Douai, northern France, where he pioneered the use of metal in airframe construction. Brothers Edouard and Charles de Nieuport set up a company specializing in high-performance monoplanes which outlived its founders, who died in flying accidents. Designer Raymond Saulnier, who had worked for Blériot, joined the Morane brothers to create Morane-Saulnier.

Germany and Britain were the only other European countries to make significant steps toward building up an aviation industry before 1914, although many of their aircraft were copies of French models, and the British used almost exclusively French aero-engines. In Germany, companies such as Albatros, Rumpler, Aviatik, and Fokker were founded; while in Britain, famous names beginning to emerge included Short Brothers, Avro, Handley Page, Sopwith, and Bristol. In the United States, Wright and Curtiss, locked in their bitter patent dispute, remained the unchallenged market leaders. Between 1910 and 1914 the United States fell behind Europe in aviation and would not

